GR AND FORGETFUL FUNCTORS DESCEND TO THE FILTERED DERIVED CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4572-4589, following the statement at 4552-4570.

Editorial note. No omissions: the only non-definitional check (forgetting the filtration) is carried out.

Allowed standard prerequisites (editorial).

- Universal property of the localization. lemma-universal-property-localization (same chapter, cited at ~4576). Produces the exact functors once inverting property is checked.
- Filtered acyclic cones and forgetful functor. lemma-filtered-acyclic (same chapter, cited at ~4583) and Homology, Lemma homology-lemma-filtered-acyclic (cited at ~4589). A filtered quasi-isomorphism has filtered acyclic cone; forgetting the filtration keeps acyclicity.

Source reference. Original reference 05QH: focused-seeds/geometry/S9-der-filtered-derived-functors/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05S1: sources/stacks-derived/pinned-original-source.tex, source lines 4509–4509. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05R7: sources/stacks-derived/pinned-original-source.tex, source lines 1688–1688. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *The functors $gr^p, gr, (forget\ F)$ induce canonical exact functors*

$$gr^p, (forget\ F) : DF(\mathcal{A}) \longrightarrow D(\mathcal{A})$$

and

$$gr : DF(\mathcal{A}) \longrightarrow D(Gr(\mathcal{A}))$$

which commute with the localization functors.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. This follows from the universal property of localization, see Lemma 05R7, provided we can show that a filtered quasi-isomorphism is turned into a quasi-isomorphism by each of the functors $\mathrm{gr}^p, \mathrm{gr}, (\text{forget } F)$. This is true by definition for the first two. For the last one the statement we have to do a little bit of work. Let $f : K^\bullet \to L^\bullet$ be a filtered quasi-isomorphism in $K(\mathrm{Fil}^f(\mathcal{A}))$. Choose a distinguished triangle $(K^\bullet, L^\bullet, M^\bullet, f, g, h)$ which contains f. Then $M^\bullet$ is filtered acyclic, see Lemma 05S1. Hence by the corresponding lemma for $K(\mathcal{A})$ it suffices to show that a filtered acyclic complex is an acyclic complex if we forget the filtration. This follows from Homology, Lemma 05QH. $\square$